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1. Introduction to the Internet.

- What is the Internet?

– The Internet is a global network that connects millions of computers and devices, enabling people to communicate, share information, and access digital resources worldwide.

- Internet vs World Wide Web.

– The WWW, or World Wide Web, is a set of software tools running on the Internet. The Internet relies on physical infrastructure, including subsea fiber-optic cables, network routers, and wireless transmission technologies (such as Wi-Fi and cellular networks) to transmit data.

– The Internet is a global network connecting computers all around the world. Therefore, the Internet is not a physical object; it is a communication network running on top of physical hardware. Beyond the World Wide Web, the Internet supports various underlying protocols and services, including Email (SMTP), File Transfer (FTP), and real-time communication platforms.

2. Core Networking Concepts

A computer network is a system of interconnected computing devices that communicate with one another to share data, services, and hardware resources. It enables high-speed communication and facilitates key network services such as email, file sharing, and Internet connectivity.

- Clients and Servers

A client refers to hardware or a software application (such as web browsers, laptops, or mobile devices) that sends requests to a server for resources or services.

A server is a central system or application (such as a web server or database server) that processes requests and delivers resources or services to clients.

- IP Addresses

An IP address is a unique numerical label assigned to every device connected to the Internet.
It allows network devices to locate and communicate with one another to transfer data accurately across local networks or the Internet.

IP Address Versions:

- ❖ IPv4: The older version of IP addressing that uses 32-bit addresses, allowing around 4.3 billion unique addresses.
- ❖ IPv6: The newer version introduced to replace IPv4 due to the high number of connected devices. It uses 128-bit addresses and provides a massive number of unique addresses.

Types of IP Addresses:

- ❖ Public IP: Assigned by the ISP (Internet Service Provider) to identify your network on the global Internet.
- ❖ Private IP: Assigned by your local router to identify internal devices within your home or office network.
- ❖ Static IP: A fixed address configured manually or provided by an ISP that does not change (commonly used for servers).
- ❖ Dynamic IP: A temporary address assigned automatically by a router using DHCP whenever a device connects to the network.

- MAC Address

- Definition: A MAC (Media Access Control) address is a physical address given permanently to a network interface card (NIC) of a computer, mobile phone, or router by its manufacturer. This address is permanently burned into the device hardware.

- Format: It is made of 12 hexadecimal digits (a combination of numbers and letters)
  Ex:- 00:1A:2B:3C:4D:5E

☐ Difference Between IP and MAC Address:

We can understand the difference between an IP address and a MAC address using an example:

- ★ IP Address: Like our home address (it can change if we move to a different network).

★ MAC Address: Like our National Identity Card (NIC) number (a unique number given at birth that never changes).

Structure of a MAC Address:
– A MAC Address has 12 hexadecimal characters (numbers 0-9 and letters A-F). It is divided into two parts:
Example: 00:1A:2B:3C:4D:5E

- ❖ First 3 parts (00:1A:2B): OUI (Organizationally Unique Identifier) - Identifies the device manufacturer (e.g., Apple, Samsung, Dell).
- ❖ Last 3 parts (3C:4D:5E): Device Identifier - A unique serial number assigned by that company for that specific device.

Types of MAC Addresses (3 Types):

- ❖ Unicast: When data is sent from one device to only one specific device.
- ❖ Multicast: When data is sent to a selected group of devices on the network.
- ❖ Broadcast: When data is sent to all devices on the network. (e.g., FF:FF:FF:FF:FF:FF)

3. The DNS Resolution Process (In-Depth)

- What is DNS?
  - ❖ It is easy for us to remember domain names like google.com. Computers only work with IP Addresses like 142.250.543.36.
    DNS (Domain Name System) acts like the phonebook of the Internet. It translates domain names like google.com into computer-friendly IP addresses.

- Step-by-Step DNS Lookup
  - ❖ When you type a domain name like google.com into a web browser, the computer follows a 5-step process to find its IP address.

1. Browser Cache & OS Cache:

The computer first checks its local memory. It checks the browser cache, and if it is not found there, it checks the Operating System (OS) cache to see if the IP address was saved recently. If found, the website opens immediately.

2. Recursive Resolver (ISP DNS):

If the IP address is not in the cache, the computer sends a request to a Recursive Resolver (provided by your ISP, like Dialog or SLT). The resolver acts as an assistant to query other servers on your behalf.

3. Root Nameserver:

If the resolver does not have the IP address, it asks the Root Server. The Root Server does not know the exact IP address, but it directs the resolver to the correct Top-Level Domain (TLD) server based on the extension (e.g., .com, .org, .lk)

4. TLD (Top-Level Domain) Nameserver:

The TLD server manages specific domain extensions (like .com). When queried, it tells the resolver where to find the Authoritative Server that holds the domain's real IP address.

5. Authoritative Nameserver:

This is the final destination and official DNS server for the website. It gives the exact IP address (e.g., 142.250.190.46) back to the resolver, which then passes it to your browser to load the website.

Summary of DNS Resolution:

Once the browser receives the correct IP address from the Authoritative Nameserver, the DNS lookup is complete, and the computer can now establish a connection to transfer data.

4. Network Protocols & Data Transfer

Network protocols are sets of rules and standards that allow different computers and devices to communicate and exchange data smoothly over a network. Data transfer is the process of breaking down information into smaller parts and sending them securely from a source device to a destination device using these protocols.

- The TCP/IP Model & OSI Model

Network models provide a structured way to understand how data moves from one device to another across a network.

OSI Model (Open Systems Interconnection).

Developed by ISO, the OSI model is a theoretical framework with 7 layers, used mainly to learn and understand how network communication works:

- ❖ Application Layer: Interacts directly with user software (e.g., Web Browsers, Email Clients).
- ❖ Presentation Layer: Handles data formatting, encryption, and decryption.

- ❖ Session Layer: Manages and maintains active connections between devices.
- ❖ Transport Layer: Ensures reliable data delivery across the network (using protocols like TCP and UDP).
- ❖ Network Layer: Handles path selection and data routing using IP addresses.
- ❖ Data Link Layer: Manages data transfer between nodes on the same local network using MAC addresses.
- ❖ Physical Layer: Transmits raw data bits (0s and 1s) over physical media like cables or Wi-Fi signals.

TCP/IP Model

Unlike the OSI model, the TCP/IP model is the practical framework used in the real Internet. It simplifies the process into 4 core layers:

- ❖ Application Layer: Combines OSI's top three layers (Application, Presentation, and Session) to run protocols like HTTP, DNS, and FTP.
- ❖ Transport Layer: Controls data packet splitting and delivery (TCP/UDP).
- ❖ Internet Layer: Routes data packets across networks using IP addresses.
- ❖ Network Access Layer: Combines OSI's Data Link and Physical layers to handle physical hardware and local transfers (Ethernet, Wi-Fi, MAC addresses).

● TCP 3-Way Handshake

Before data can be transferred between a client and a server, TCP establishes a reliable connection using a 3-step process known as the 3-Way Handshake. (Think of it like a quick check before starting a phone call to make sure both people can hear each other clearly).

- ❖ SYN (Synchronize): The client (your computer) sends a SYN packet to the server to ask for permission to initiate a connection.
- ❖ SYN-ACK (Synchronize-Acknowledge): Once the server gets the request, it replies with a SYN-ACK packet. This confirms that it received the client's request (ACK) and that it is also ready to establish the connection (SYN).
- ❖ ACK (Acknowledge): The client receives the SYN-ACK packet and sends back a final ACK packet to confirm. At this point, the connection is successfully established, and data transfer can begin.

- Data Packets & Routings

Once a connection is established, data is transferred across the network using packets and routing.

Data Packets

Large files (like images, videos, or web pages) aren’t sent over the internet as a single block. Instead, they are broken down into smaller chunks called Data Packets for efficient transmission.

Each packet contains two main parts:

- Header: Contains metadata such as the source IP address, destination IP address, and sequence numbers (to help rebuild the file in the correct order).
- Payload: The actual raw data being transmitted.

Routing

Routing is the process of directing these data packets from the sender to the receiver across multiple networks.

- Role of Routers: Routers inspect the destination IP address of each incoming packet to determine where it needs to go.
- Best Path Selection: Routers evaluate network traffic to select the fastest and most efficient route for the packet.
- Reassembly: Because packets can travel through different paths, they may arrive out of order. The receiving device uses the sequence numbers in the header to reassemble all the packets back into the original file.

5. What Happens When You Type a URL in a Browser?

When you enter a web address like https://www.google.com into your browser, a multi-step background process takes place to find the server and load the webpage:

1. Entering the URL

You type the address into the browser's address bar. The browser breaks down the URL to identify the protocol (https://) and the domain name (google.com).

2. DNS Lookup

The browser needs to find the server's IP address. It checks local caches first, and if not found, queries DNS servers to translate google.com into a numerical IP address (e.g., 142.250.190.46).

3. Establishing TCP Connection (3-Way Handshake)

Once the IP address is found, the browser initiates a reliable connection with the target server using the TCP 3-Way Handshake (SYN → SYN-ACK → ACK).

4. TLS/SSL Handshake (HTTPS Security)

Since the URL uses HTTPS, an additional security step occurs. The browser and server exchange cryptographic keys to encrypt the connection and keep data transfers secure.

5. Sending HTTP/HTTPS Request

With a secure connection established, the browser sends an HTTP/HTTPS request (such as a GET request) asking the server to send over the webpage files.

6. Server Response & Rendering the Webpage

The server processes the request and sends back the webpage content (HTML, CSS, JavaScript, images) as data packets. The browser processes these files and renders the final webpage on your screen.

6. Web Security Basics

Web security protocols ensure that data transferred between a user's browser and a server remains private and protected from tampering.

- HTTP vs HTTPS

❖ HTTP (Hypertext Transfer Protocol): Transmits data in plain text. This means any third party intercepting the network connection can easily read sensitive information like passwords or credit card numbers.

❖ HTTPS (HTTP Secure): Encrypts the data transferred between the browser and the server. Even if an attacker intercepts the data packets, they only see unreadable ciphertext instead of plain text.

- SSL/TLS Encryption

HTTPS uses SSL/TLS protocols to secure communications. While SSL is the older version, modern web security relies on TLS (Transport Layer Security).

- ❖ Asymmetric Encryption: Uses a pair of keys (a public key to encrypt and a private key to decrypt) during the initial connection setup to securely verify identity.
- ❖ Symmetric Encryption: Once the secure connection is verified, both sides switch to a single shared key (session key) to quickly encrypt and decrypt data during the rest of the session.

7. Conclusion

Computer networking is the backbone of the modern Internet, enabling seamless communication between devices across the globe. Understanding core fundamentals - such as MAC and IP addressing, network hardware, DNS resolution, and the TCP/IP suite-provides a clear picture of how data is structured and routed efficiently.

Furthermore, combining connection handshakes with encryption protocols like TLS/SSL ensures that web browsing remains both reliable and secure. Mastering these core concepts builds a strong foundation for exploring advanced topics in software engineering, system architecture, and web development.

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